Introduction to DevOps

Definition of Devops:

DevOps is a culture and set of practices that improves collaboration between developers and operations teams to build, test, and release software quickly and efficiently.

DevOps is a combination of:

- **Development (Dev)**
- **Operations (Ops)**

It is a **process/methodology** that helps teams:

- Build software faster
- Deliver updates quickly
- Improve collaboration

Goal of DevOps

- Faster delivery
- High quality software
- Automation
- Continuous improvement

Traditional vs Devops

Traditional Model

- Dev & Ops teams work separately
- Slow releases
- More errors

DevOps Model

- Dev & Ops work together
- Fast releases
- Automated processes

DevOps Lifecycle

1. **Planning**
2. **Development**
3. **Build**
4. **Testing**
5. **Release**
6. **Deploy**
7. **Operate**
8. **Monitor**

This cycle repeats continuously.

DevOps Tools (Basic)

- **Version Control** → Git
- **CI/CD** → Jenkins
- **Build Tools** → Maven, Gradle
- **Configuration Mgmt** → Ansible
- **Containerization** → Docker
- **Orchestration** → Kubernetes
- **Cloud** → AWS, Azure

What is CI/CD?

CI (Continuous Integration)

- Developers push code frequently
- Code is automatically built & tested

CD (Continuous Delivery/Deployment)

- Code is automatically deployed to servers

Importance of Automation

- Reduces manual work
- Saves time
- Avoids human errors

DevOps + Cloud

DevOps works best with cloud platforms like:

- AWS
- Azure
- Google Cloud

Cloud provides:

- Scalability
- Flexibility
- Pay-as-you-use

Key Concepts

- Infrastructure as Code (IaC) → Terraform
- Monitoring → Prometheus, Grafana
- Logging → ELK Stack

Benefits of DevOps

- Faster delivery
- Better collaboration
- Reliable systems
- Continuous feedback

Real-Time Example

When you use apps like:

- Instagram
- WhatsApp

Updates come frequently without downtime → That’s Devops